Lochan Surya Teja Neeli

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.83
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2019	97.30%
Jawahar Navodaya Vidyalaya	Class X (Central Board of Secondary Education)	2017	10.00%

SKILLS

Programming Languages: C, C++, Python

Software Tools: UNIX/Linux-based Operating System, gcc, make, Bash and Command Line Interface, Vim, CMake, LaTeX, git

Programming Paradigms: Structured Programming, Object-Oriented Programming

Theory: Computer Science, Discrete Probability, Mathematical Modelling, Mathematical Analysis, Algorithm Design

PROJECTS

Extending BANG for Dynamic Nearest Neighbor Queries(CS5013: GPU Programming)

- A full GPU-accelerated implementation of FreshVamana. A **high-throughput search(64000 QPS), supporting dynamic insertions, deletions, and periodic graph consolidation** using a CUDA-optimized Vamana graph.
- Implemented a worklist-based graph algorithm for approximate nearest neighbors search.
- Skills: **C Programming Language**, **CUDA, SIMT, Systems Software Design, Data Structuring for GPU-based Algorithms**

Odd-Even Brick Sort on GPU using Warp-Synchronous Programming(CS5013: GPU Programming)

- Implemented a SIMT warp-based CUDA implementation for optimizing the Sorting sub-routine on NVIDIA GPUs.
- Achieved the **best running-time (1.41ms) for the almost-sorted case** in the GPU Sorting Hackathon.
- Applications: **Accelerating On-Line Vector DataBase Search for Dynamic Queries**
- Skills: Introduction to CUDA Warp-Synchronous programming, C Programming

PolyFeat Cost Model for Loop Optimizations(CS5260: Compiler Optimizations)

- Experimented and Analyzed PolyFeat Analyzer's **Cost Model for Operational Intensity(OI)** Modelling for Loop Optimizations.
- Reported the Limitations of the Cost Model for estimating the Tiled Matrix Multiplication cache misses on my Personal Linux Laptop.
- Tools Used: pocc(polyhedral compilation), gcc, Ubuntu-24.04(LTS), Dinero-IV Cache Simulator
- Skills: **Cache Modelling**, **Introduction to Polyhedral Compilation for Parallel Code Generation, Precise Instrumentation of Code, Data Consolidation and Reporting for Systems Software, Conditional Compilation, Source Build**

Binary Image Classification with CBAM-Enhanced ResNet(CS5480: Deep Learning)

- Designed a ResNet-34-like CNN from scratch with **Convolutional Block Attention(CBAM), GeM pooling, and stochastic depth (DropPath)** for binary image classification.
- Developed a **shape-aware edge-map preprocessing pipeline** geometric features and eliminate spurious colour cues.
- Built an end-to-end training pipeline with AdamW, cosine learning-rate scheduling, mixed precision, TTA, and threshold calibration, achieving **77.53% leaderboard accuracy**, a **1.9% improvement** over the previous best.

- Skills: Python, PyTorch, TorchVision, AI Training, Data Pre-Processing Techniques

Computer Vision--Segmentation, Object Detection & Vision Transformers(CS5290: Computer Vision)

- **Implemented U-Net-based binary image segmentation**, comparing architectures with and without skip connections and achieving **0.9364 Dice** and **0.8822 IoU scores** with skip connections.
- **Trained YOLO11n for pedestrian detection** on the Penn-Fudan dataset, including mask-to-bounding-box conversion, YOLO-format dataset preparation, confidence-threshold analysis, and NMS, achieving **0.9427 mAP@0.5**.
- **Implemented a Vision Transformer from scratch** with patch embedding, positional embeddings, multi-head self-attention, and transformer blocks; trained a 6-layer ViT on CIFAR-10 achieving **79.51% test accuracy** and analyzed attention maps and attention-distance across layers.

Contrastive Representation Learning for Image Classification[Github]

- **Implemented a VGG-16-based image encoder** with batch normalization and a 64-dimensional embedding space for learning compact visual representations.
- **Trained the encoder using triplet margin loss** with batch-wise anchor, positive, and negative sampling, and analyzed the learned representations using **t-SNE** and contrastive-loss curves.
- **Fine-tuned classifiers on the learned embeddings**, experimenting with linear and neural-network classifier heads and early stopping; achieved **86.35% test accuracy** with the 64-dimensional representation.

KEY COURSEWORK

GPU Programming, Computational Number Theory and Algebra, Compiler Optimizations, Algorithmic Techniques for Large Graphs, Approximation Algorithms, Computer Vision, Deep Learning

CERTIFICATIONS

- Code In Place 2026 : Stanford University: Introduction to Programming in Python: Console Programming & Graphics Programming
- NPTEL: Mathematical Foundations of Machine Learning**(81%),** Introduction to Graph Algorithms**(94%)**, Intermediary Algorithm*, Programming in Modern C++*

POSITIONS OF RESPONSIBILITY

Teaching Assistantship

- Introduction to Programming(Jul 2025 - Nov 2025, Jul 2026 - Nov 2026)
- Operating Systems II (Jan 2026 - Apr 2026)

EXTRACURRICULAR

chess.com/lochan80: Solved 68, 686 puzzles with the best puzzle rated at 1973

HOBBIES

Reading Programming Languages' Documentation, CppCon Talks, Programming, Debugging (GDB), File Systems, Assembler, x86 ISA, C, C++, Python, Rust, Scheme--*SICP,* Build Systems, GNU Autotools, Systems Software: Architecture, Design, and Integration.